Lesson plan · for the teacher

Relationships

Data science · Level 2 · Lesson 3

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Draw and read a scatter plot.
- Read a correlation r.
- Ask what else could explain a pattern.

Activity: A class scatter chart

1. Warm-up (5 min): هل يسير النوم والدرجات معًا؟
2. Teach (8 min): Sketch rising, falling and cloud patterns with their r. Tell the ice cream story.
3. Plot (15 min): Pupils measure in pairs and add one dot each to axes on the board.
4. Discuss (7 min): Is the pattern strong? Then read two school claims and ask what else could explain each.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Why is height linked to arm span?
- What could explain sleep and marks going together?
- How would you test a cause fairly?

Exit ticket

- Sketch a scatter plot with r close to -0.9.

Try it in class tomorrow
Pupils measure their height and arm span and add one dot each to a scatter chart on the board. Is the pattern strong? Then discuss a pattern that is not a cause.

Worksheet · for pupils

Relationships

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Name _____ Class _____ Date _____

1 **More books at home goes with higher marks (r = 0.6). Does that prove books cause higher marks?**

◯ Yes ◯ No: something else may explain both ◯ Only if r is 1

2 **Match each pattern to its correlation.**

Dots close to a rising line	• • • •	About -0.9
A shapeless cloud	• • • •	About 0
Dots close to a falling line	• • • •	About 0.9

3 **Finish the program so it prints the correlation r between study hours and marks, rounded to 2 decimal places.**

```
hours = [1, 2, 3, 4, 5]
marks = [8, 9, 11, 12, 15]
mx = sum(hours) / 5
my = sum(marks) / 5
sxy = 0
sxx = 0
syy = 0
for i in range(5):
    sxy = sxy + (hours[i] - mx) * (marks[i] - my)
print(sxy)
```

Write your program:

4 **Sleep and marks go together. Which could also explain it? Choose all that apply.**

☐ A calm home with help for homework ☐ Good health ☐ Living close to school

☐ The colour of the school uniform

5 **One unusual pupil can change the correlation a lot.**

◯ True ◯ False

Which school "fact" have you heard that might be a correlation, not a cause?